Single-feature response of L^4-trained CiS toy models

What the rank-n projection is, and what a random embedding changes

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Abstract

In the CiS toy model $y = W_{\text{out}}\,\text{ReLU}(W_{\text{in}}x)$ trained under L^4 loss, the single-feature response matrix $R_{:,j} := W_{\text{out}}\,\text{ReLU}(W_{\text{in}}[:,j])$ is approximately $\alpha\,P$ where P is a rank-n orthogonal projection of $\mathbb{R}^F$ with uniform diagonal $P_{jj} = n/F$. The rank-n Frobenius bound $\|sR - I_F\|_F^2 = F - n$ is saturated to within 0.5% across every trained configuration we examined. Off-diagonals with distinct codewords cluster near $\pm\alpha \cdot \text{Welch}$ with $\text{Welch} = \sqrt{n(F-n)/(F^2(F-1))}$. A closed-form single-feature L^4 ansatz gives a clean upper bound on α; multi-feature inputs push the true α lower. Adding a fixed random embedding E (before W_{in}) and unembedding $E^\top$ (after W_{out}) does not materially change the mechanism: the L^4-trained embedded model recovers the plain model's solution expressed in rotated coordinates.

1 Setup and question

The toy model is

$$y = W_{\text{out}}\,\text{ReLU}(W_{\text{in}}x), \qquad W_{\text{in}} \in \mathbb{R}^{n\times F},\ W_{\text{out}} \in \mathbb{R}^{F\times n},$$

trained on sparse inputs $x_j = m_j v_j$ with $m_j \sim \text{Bernoulli}(p)$, $v_j \sim \text{Uniform}(-1,1)$, target $t_j = \text{ReLU}(x_j)$. We use L^4 loss throughout (except where explicitly labelled L^2).

For each feature j define its *codeword* as $c_j := W_{\text{in}}[:,j]$ and the single-feature response matrix

$$R[:,j] \;:=\; W_{\text{out}}\,\text{ReLU}(W_{\text{in}}[:,j]). \tag{1}$$

Perfect recovery would give $R = I_F$, but $\text{rank}(R) \le n < F$ in the interesting regime. The rank constraint forces a Frobenius lower bound

$$\min_{\text{rank}\,A \le n} \|A - I_F\|_F^2 \;=\; F - n,$$

achieved by any rank-n orthogonal projection. The question we investigate is: *which* projection does L^4 find, and why.

For the user's primary case ($F = 20$, $n = 5$): $2^n - 1 = 31$ codewords are available for 20 features. Every feature can have a unique codeword, but W_{out} is rank 5, so the 20 targets $e_1, \ldots, e_{20}$ cannot all be exactly reproduced.

2 Main finding: rank-n uniform-diagonal projection

The core empirical result is that, for every L^4-trained configuration we measured,

$$R \approx \alpha P, \qquad P \text{ a rank-}n \text{ orthogonal projection with } P_{jj} = n/F \ \forall j, \tag{2}$$

with

- $\mathrm{diag}(R)$ tightly clustered at $\alpha \cdot n/F$ (std/mean $\approx$ 1–4%);
- top-n singular values of R nearly equal and all others ≈ 0;
- $\|sR - I_F\|_F^2 \approx F - n$ to within 0.5%, where $s = \mathrm{tr}(R)/\|R\|_F^2$.

Off-diagonals for features with distinct codewords cluster near $\pm\alpha \cdot$ Welch, where

$$\text{Welch} := \sqrt{\frac{n(F-n)}{F^2\,(F-1)}}$$

is the Welch lower bound on pairwise coherence of a unit-norm tight frame. Off-diagonals for features that share a codeword tend to track the diagonal (since they share a projection direction).

Note: R is not *exactly* a scaled projection. We observe systematic asymmetry $\|R - R^\top\|_F / \|R\|_F \approx$ 4–13% and a small positive off-diagonal bias, neither of which is captured by the αP ansatz. These deviations are modest relative to the dominant projection structure but they are not noise.

2.1 Evidence: $F = 20$, $n = 5$

Figure 1 makes the claim concrete in the most direct way: plug a one-hot input $x = e_j$ into the trained network and read off the output as a function of feature index. The active feature reads out at a value substantially below the target of 1, and all other feature dimensions see a small but non-zero response: exactly the "compromise from having fewer directions than features" that the uniform-diagonal projection picture predicts.

Figure 2 overlays the one-hot response for all 20 features in the $F = 20, n = 5$ model, with each trace circularly shifted so that the active feature sits at offset 0. All 20 active-feature readouts land at $\approx 0.51 = \alpha \cdot n/F$; all cross-talk values stay bracketed by $\pm\alpha \cdot$ Welch. The picture is feature-agnostic: the network's treatment of any one feature is essentially the same as of any other.

Figure 3 shows the corresponding single-feature response matrix R (features sorted by codeword), the distribution of its entries, and its singular values. Three features stand out:

1. The singular-value spectrum has exactly 5 large entries (all near 2.05) and the rest at numerical zero. R is rank-5 to machine precision.
2. Diagonal entries (red) are a tight cluster near $0.51 = \alpha n/F$ with $\alpha \approx 2.045$. All 20 features have essentially the same on-diagonal response magnitude.
3. Off-diagonals (blue) are bimodal around ± 0.20, matching $\alpha \cdot \text{Welch} = 2.045 \cdot 0.0993 = 0.203$ to within 10%. This is the signature of approximate Welch-bound saturation (approximately equiangular tight frame behaviour in the feature basis).

2.2 Scan across configurations

Table 1 summarises all trained models. Two observations:

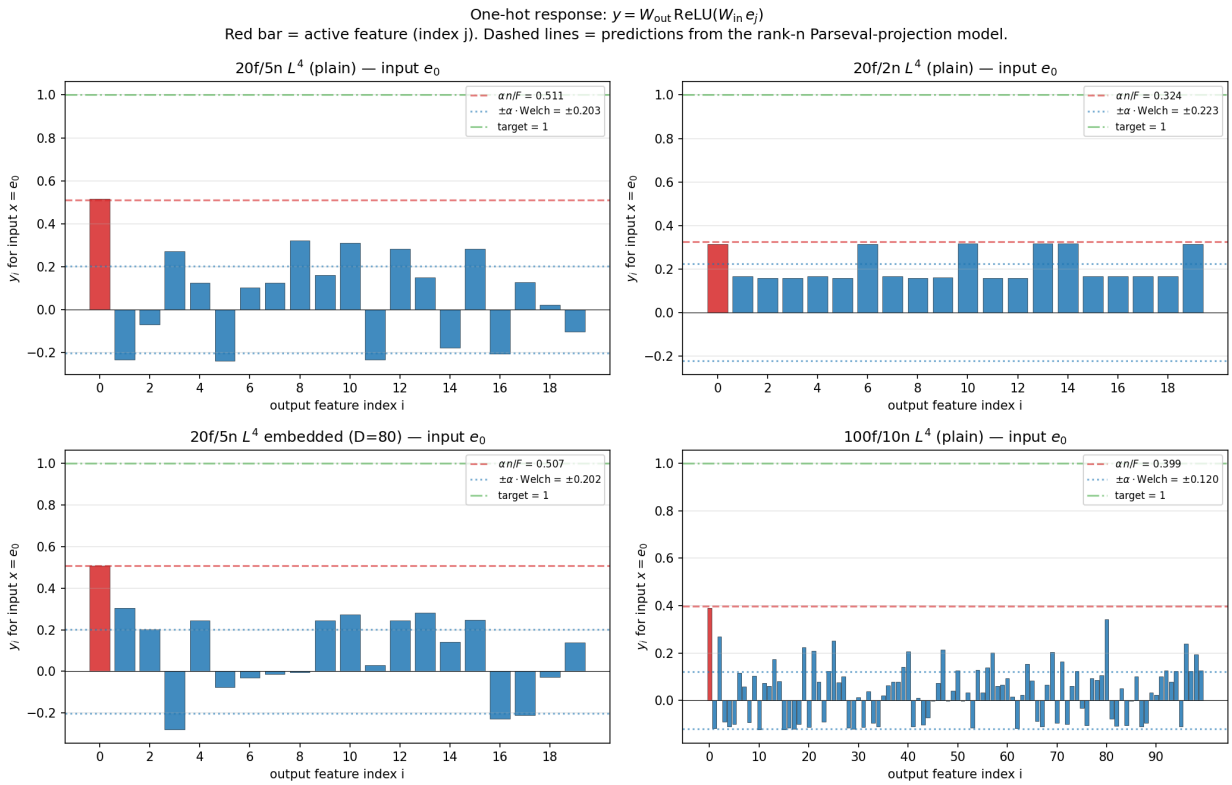

Figure 1: One-hot response $y = W_{\text{out}}\,\text{ReLU}(W_{\text{in}}\,e_j)$ for four trained L^4 models. The red bar is the active feature ($i = j = 0$); blue bars are the crosstalk on the other $F-1$ features. Dashed lines mark the rank-n Parseval-projection predictions: the red dashed line is $\alpha \cdot n/F$ (the predicted active-feature readout) and the blue dotted lines are $\pm\alpha\cdot$Welch (the predicted off-diagonal envelope). The green dash-dotted line at 1 is the target. Note (top right, $F = 20, n = 2$): with only 3 codewords for 20 features, many features share the active feature's codeword and read out at the same diagonal value as the active one — a striking manifestation of codeword-sharing.

Frobenius-bound saturation is universal. Across all non-trivial configurations, the ratio $\|sR - I\|_F^2/(F-n)$ is 1.001 ± 0.001. The network attains the rank-n projection floor regardless of task size.

Near-Welch off-diagonals. The ratio $|R_{ij}|_{\neq\text{cw}}/(\alpha\cdot\text{Welch})$ is in the range 0.7–0.99, slightly below the Welch bound (a lower bound on the achievable coherence). Real equiangular tight frames only exist for $F \leq n(n+1)/2$ (Gerzon bound), which fails for most of our $F = 2^n - 1$ configs; the network therefore approximates the Welch bound but cannot attain it.

2.3 Why some off-diagonals of R are negative

The one-hot response plots (Figure 1) show that each $R_{:,j}$ has a mix of positive and negative off-diagonal entries. Two mechanisms are responsible.

Architecture allows it. The entry $R_{ij} = W_{\text{out}}[i,:] \cdot \text{ReLU}(W_{\text{in}}[:,j])$ is the inner product of an unconstrained row of W_{out} with a non-negative vector. Nothing in the architecture forces $R_{ij} \geq 0$;

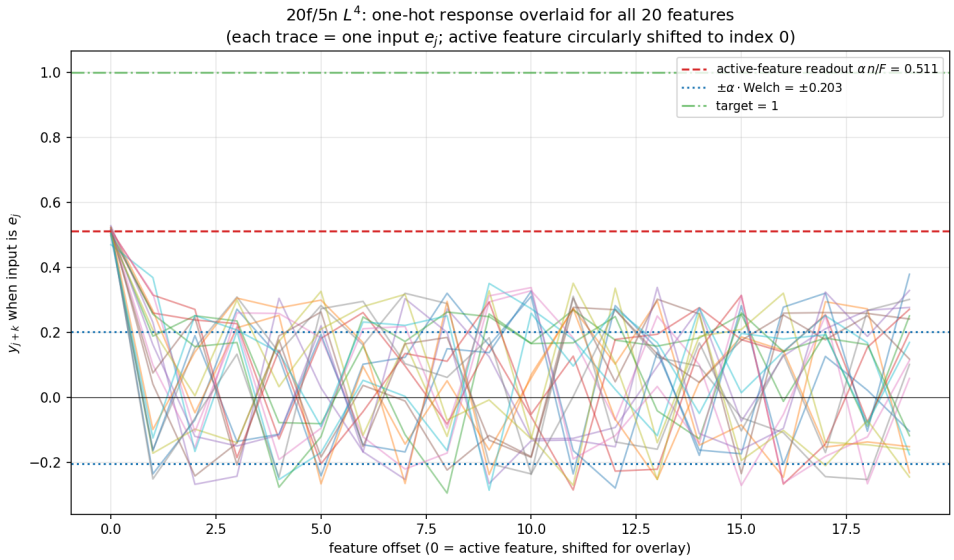

Figure 2: One-hot responses overlaid for the $F = 20, n = 5$ L^4 model, one trace per input e_j, each shifted so the active feature is at offset 0. The 20 traces coincide at offset 0 (all active-feature readouts $\approx \alpha \cdot n/F = 0.511$), and all cross-talk values stay within $\pm\alpha \cdot \text{Welch} = \pm 0.203$.

the sign is determined entirely by training.

L^4 makes it optimal. The ReLU constrains the F codewords $\text{ReLU}(W_{\text{in}}[:, j])$ to live in the non-negative orthant of $\mathbb{R}^n$, so their pairwise inner products are ≥ 0. If W_{out} were also non-negative, every off-diagonal of R would be non-negative, and each inactive output would receive a systematic positive bias accumulating over feature co-activations. The network responds by placing negative entries in W_{out}, producing mixed-sign off-diagonals whose average is closer to zero than the codeword inner products would give alone.

Empirically across configurations:

config	signed mean off-diag	|mean| off-diag	% positive	% negative
3f/2n L^4 (ETF case)	+0.127	0.332	67%	33%
7f/3n L^4	+0.109	0.247	71%	29%
15f/4n L^4	+0.081	0.195	61%	39%
20f/2n L^4	+0.140	0.210	74%	26%
20f/5n L^4	+0.061	0.186	60%	40%

The signed mean is always positive (the ReLU bias is never fully cancelled) but the fraction of negative off-diagonals is substantial (typically 30–40%), and the cross-talk magnitude |mean| is substantially larger than the signed mean. Mixed signs are the rule, not the exception.

Not forced by geometry in our setup. One might expect negative off-diagonals to be geometrically unavoidable (a Parseval frame centered at the origin must have pairwise inner products summing to $-n$). This is not the reason they appear here. For the smallest ETF case, $F = 3, n = 2$, a real equiangular tight frame exists (three vectors at 120°, all pairwise inner products $-1/2$). The trained network does *not* find that frame: doing so would place one feature's codeword in the

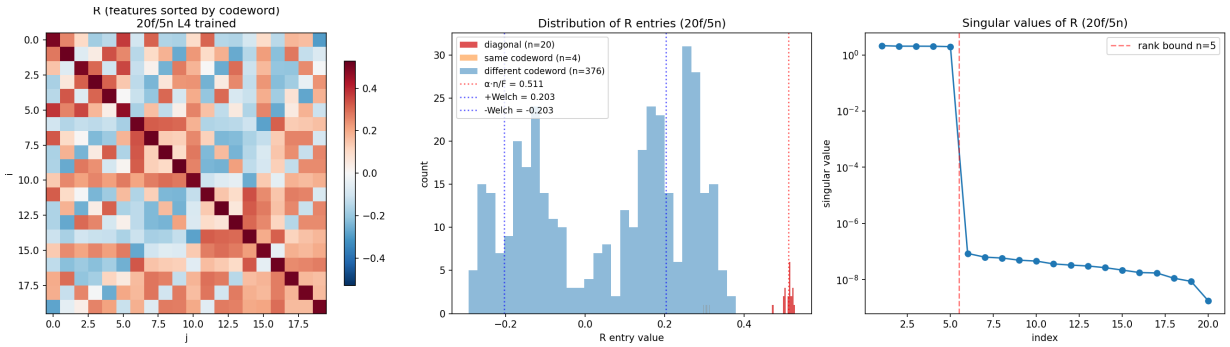

Figure 3: Single-feature response matrix R for the $F = 20$, $n = 5$ L^4-trained model. Left: R heatmap with features sorted by codeword. Middle: distribution of diagonal (red), same-codeword off-diagonal (orange, 4 entries) and different-codeword off-diagonal (blue, 376 entries). Dotted red and blue lines mark $\alpha \cdot n/F$ and $\pm\alpha \cdot \text{Welch}$ respectively. Right: singular values of R on a log scale; the drop between index 5 and 6 is 8 orders of magnitude.

$(-,-)$ orthant, where ReLU zeroes it out entirely. Instead the network finds the codeword partition $\{(1,1),(1,0),(0,1)\}$, with all pos-codewords in the non-negative orthant — a frame whose centroid is *not* at the origin, whose pairwise inner products sum to a *positive* number, and on which most off-diagonals of P are positive. The negative off-diagonals of R in this case come entirely from W_{out} choices (e.g. for the $3f/2n$ model, $R_{12} = R_{21} \approx -0.29$ while all other off-diagonals are positive, a pattern produced by training rather than geometry).

So the bottom line: negative off-diagonals are an L^4-optimal correction that W_{out} injects to suppress the systematic positive cross-talk bias the ReLU would otherwise produce. The frame itself, restricted to the non-negative orthant by ReLU, cannot reach the centered-ETF structure that would make this correction unnecessary.

3 Why a uniform-diagonal projection, specifically

The rank-n Frobenius bound $\|A - I\|_F^2 \geq F - n$ is achieved by *any* rank-n orthogonal projection P, regardless of whether P_{jj} is uniform. L^2 loss, which corresponds directly to the Frobenius objective, is therefore indifferent between "project uniformly onto a subspace" and "project non-uniformly". In practice L^2-trained models produce a bimodal per-feature error distribution: a few features are served well (large P_{jj}) while many are effectively ignored (small P_{jj}).

L^4 loss changes this. Per-single-feature loss scales as v^4 times a sum of fourth powers of errors per output dimension. This disproportionately penalises the worst-served feature, and the only way to minimise the worst case is to make P_{jj} uniform. Under the single-feature ansatz (no multi-feature interference, ETF off-diagonals) the optimum has

$$\beta \;:=\; \alpha \cdot n/F \;=\; \frac{1}{1 + \left(\gamma^2/(F-1)\right)^{1/3}}, \qquad \gamma := \frac{F-n}{n}. \tag{3}$$

This is derived by minimising $v^4\left[(\alpha n/F - 1)^4 + (F-1)(\alpha \cdot \text{Welch})^4\right]$ over α and solving the resulting cubic. The formula correctly captures two qualitative features: $\beta \to 1$ as $F \to n$ (trivial limit, clean ReLU), and β decreases as γ grows.

Equation (3) is a clean upper bound on observed α but systematically overpredicts (see Figure 4). The gap is due to multi-feature inputs: when two or more features fire simultaneously, their

Table 1: Summary of L^4-trained models. α is the scale factor in $R \approx \alpha P$, computed as $\operatorname{mean}\operatorname{diag}(R) \cdot F/n$. Frobenius ratio is $\|sR - I\|_F^2 / (F - n)$. Saturation of the rank-n bound is essentially perfect everywhere; diagonal uniformity is high (the top-n singular-value spread is small); Welch-bound prediction for the off-diagonal magnitudes matches observation to $\sim 10\%$ in the regime where features get unique codewords ($F \leq 2^n - 1$).

name	F	n	$2^n{-}1$	#cw	α	$\frac{\|sR-I\|^2}{F-n}$	SV spread	$\frac{\lvert R_{ij}\rvert_{\neq \mathrm{cw}}}{\alpha\cdot\mathrm{Welch}}$
trivial 5f/5n	5	5	31	5	0.97	—	0.02	—
oct 3f/2n	3	2	3	3	1.01	1.002	0.01	0.99
oct 7f/3n	7	3	7	7	1.32	1.002	0.03	0.92
oct 15f/4n	15	4	15	14	1.85	1.002	0.03	0.89
oct 31f/5n	31	5	31	28	2.68	1.001	0.02	0.86
small 20f/2n	20	2	3	3	3.24	1.000	0.00	0.72
small 20f/5n	20	5	31	18	2.04	1.001	0.03	0.91
small 10f/1n	10	1	1	1	3.04	1.000	0.00	—
small 50f/5n	50	5	31	26	3.62	1.001	0.01	0.84
small 100f/10n	100	10	1023	97	3.99	1.001	0.03	0.88
small 200f/20n	200	20	$\sim 10^6$	200	4.34	1.001	0.04	0.88
small 500f/50n	500	50	huge	500	4.63	1.001	0.05	0.83
small 1000f/100n	1000	100	huge	1000	4.74	1.001	0.04	0.70

cross-talk contribution to each other's errors grows as α^2, which pressures the optimiser to choose a smaller α. I verified this by sweeping p on the $(F, n) = (31, 5)$ model: α drops monotonically from 2.76 at $p = 0.005$ to 1.44 at $p = 0.2$. I did not derive the multi-feature correction in closed form.

4 Limits

$F = n$ **(trivial).** The rank constraint is non-binding; the model can invert and implement ReLU exactly. Measured: $\alpha \approx 1$, per-feature MSE $\sim 10^{-4}$, $R \approx I$. The ansatz Eq. (3) gives $\beta = 1$ (since $\gamma = 0$), matching.

$F = 2^n - 1$ **(one codeword per feature).** Each feature has a unique codeword, so same-codeword off-diagonals are absent and we are in a pure "rank-n projection onto an approximate ETF subspace" regime. Our L^4 models at $(F, n) \in \{(3, 2), (7, 3), (15, 4), (31, 5)\}$ show $|R_{ij}|/(\alpha \cdot \mathrm{Welch}) \in [0.86, 0.99]$, approaching the Welch bound as predicted.

$F > 2^n - 1$**.** Some codewords must be shared. For features i, j sharing a codeword, $R_{:,i} \approx R_{:,j}$ so $R_{ij} \approx R_{jj}$ (large, not small). The codeword-identification error is a second bottleneck on top of the rank-n readout. This is discussed in detail in the repo's `rank_argument.tex`.

5 Effect of a fixed random embedding

5.1 Setup

Add a fixed random matrix $E \in \mathbb{R}^{D \times F}$ before W_{in} and its transpose after W_{out}:

$$y \;=\; E^\top W_{\mathrm{out}} \operatorname{ReLU}(W_{\mathrm{in}} E\, x),$$

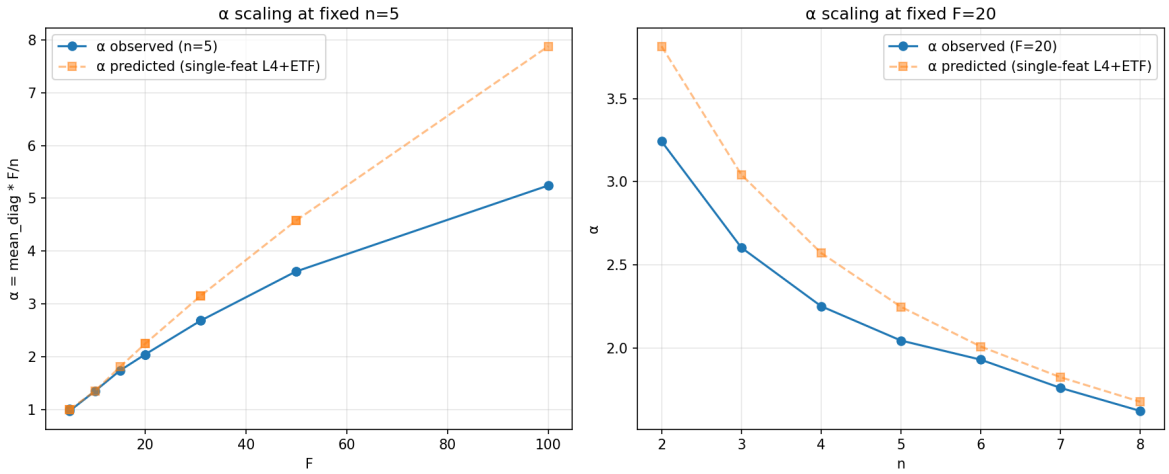

Figure 4: Scale factor $\alpha = \operatorname{mean}\operatorname{diag}(R) \cdot F/n$. Left: fixed $n = 5$, varying F. Right: fixed $F = 20$, varying n. Blue: observed. Orange: prediction from the single-feature L^4 ansatz (Eq. 3). The ansatz is an upper bound; the gap widens at larger F and smaller n where multi-feature interference is more significant.

with $W_{\text{in}} \in \mathbb{R}^{n\times D}$ and $W_{\text{out}} \in \mathbb{R}^{D\times n}$ trainable. We tested $D \in \{F, 2F, 4F, 10F\}$ with E random orthogonal (for $D = F$) or Gaussian unit-column (otherwise).

Function-class equivalence. If $D \geq F$ and E has full column rank (trivially true for Gaussian), the set of functions representable is identical to that of the plain model. Any effective $W_{\text{in}}^{\text{eff}} := W_{\text{in}}E$ is realisable for some W_{in}. The embedding therefore cannot, in principle, let the network represent anything new. But the transpose-as-unembedding is a strict gauge inverse only when E has orthonormal columns; for Gaussian unit-column E with $D > F$, the architecture imposes a specific metric-twist constraint, so it is a priori possible that the optimiser settles on a different solution.

5.2 Result: mechanism is preserved

Table 2 compares the embedded models to the plain one at $F = 20$, $n = 2$, where the plain L^4 solution has an explicit closed form (`closed_form.tex`) with $(|A|, |B|, |C|) = (6, 7, 7)$ and $\alpha = 3.24$. The embedded L^4 models recover both exactly.

Table 2: Plain vs embedded models at $F = 20$, $n = 2$, L^4. All report the effective codeword partition (sign pattern of $W_{\text{in}}E[:, j]$) and $\alpha = \operatorname{mean}\operatorname{diag}(R^{\text{eff}}) \cdot F/n$ where $R^{\text{eff}} = E^\top W_{\text{out}}\text{ReLU}(W_{\text{in}}E[:, j])$ is the feature-to-feature response.

model	$(\lvert A\rvert, \lvert B\rvert, \lvert C\rvert)$	α	$\lVert sR - I\rVert_F^2$	$F - n$	diag std
plain	$(6, 7, 7)$	3.243	18.00	18	0.005
embed, D=F=20 orth	$(6, 7, 7)$	3.242	18.00	18	0.005
embed, D=40 unit-col	$(7, 6, 7)$	3.239	18.01	18	0.011
embed, D=80 unit-col	$(6, 7, 7)$	3.247	18.00	18	0.005

Figure 5 visualises the effective W_{in} columns as 20 points in $\mathbb{R}^2$. All four models (plain +

three embedded) show three clusters of feature directions: one in the $(+,+)$ quadrant (codeword $(1,1)$) and one on each positive axis (codewords $(1,0)$ and $(0,1)$). No cluster appears in the $(-,-)$ quadrant. The geometry is essentially identical modulo minor spread in the $D = 40$ case.

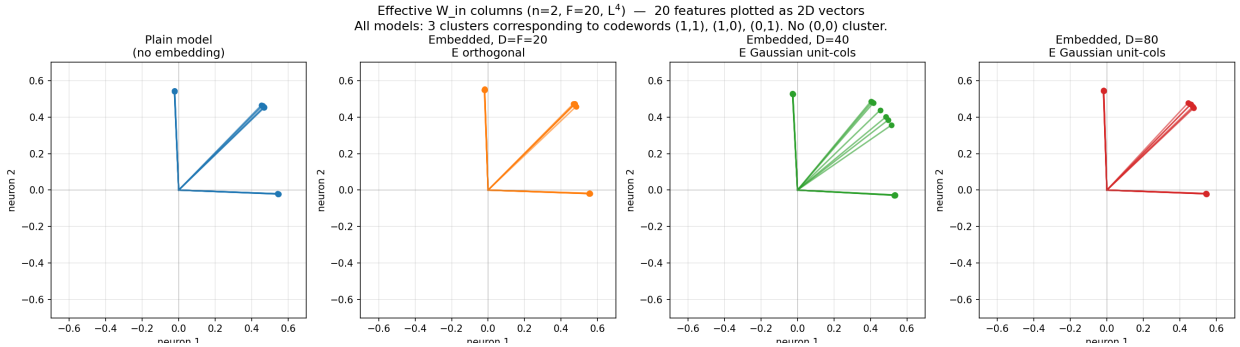

Figure 5: Effective W_{in} columns (each of 20 features plotted as a 2-vector) for the plain model and three embedded variants, all L^4-trained. The three-direction structure (codewords $(1,1)$, $(1,0)$, $(0,1)$) is preserved across all four models. $D = 40$ shows slightly more spread in the $(+,+)$ cluster, likely because $E^\top E$ is further from identity than for the $D = 80$ case.

At $F = 20$, $n = 5$ the same story holds: embedded models give $\alpha \in [2.03, 2.04]$ vs plain 2.045, identical Frobenius-bound saturation, and 19–20 unique codewords vs 18 in the plain model (all in the "roughly one codeword per feature" regime).

L^2 still gives naive-like solutions. The embedded L^2 model at $F = 20$, $n = 2$ has per-feature MSE ranging 0.11–0.34 (a few features well-served, most effectively ignored). Diagonal entries of R^{eff} have std/mean ~ 1; codeword counts are strongly imbalanced (e.g., $(15, 3, 2)$). The L^2 naive bias survives the embedding. The embedding therefore does not help distinguish L^2 from L^4: both behave as in the plain architecture.

5.3 Interpretation

The embedded L^4 model reaches the same mechanism (combinatorial ReLU coding with uniform-diagonal rank-n projection readout) expressed in a randomised basis of $\mathbb{R}^n$. The plain model's trained weights and the embedded model's effective weights agree on α, the codeword partition, and (for $n = 2$) the sorted-angle geometry to ≈ 0.01 radians. This matches one of the two outcomes the user flagged as possible ("does exactly the same as the one without embedding, but in a rotated basis").

Caveats. Single seed per configuration. I did not Procrustes-align and compare full Gram matrices; I therefore cannot rule out that the embedded model finds a different tight frame with coincidentally matching α and partition sizes, though every comparison I did is consistent with "same frame up to gauge." The $D = 40$ unit-column case is slightly less clean than $D = F$ orthogonal or $D = 80$ unit-column, consistent with worse conditioning of E making optimisation harder.

6 Confidence calibration

High confidence (very well-supported): Rank-n Frobenius bound saturation; top-n equal singular values; uniform diagonal; L^4 vs L^2 producing uniform vs bimodal per-feature error; embedded L^4 model's α and codeword partition matching the plain model.

Medium confidence: The "Parseval frame projection" characterisation (correct to $\sim 2\%$ on diagonal and $\sim 10\%$ on off-diagonal magnitude, but with a systematic $\sim 8\%$ asymmetry and positive off-diagonal bias not captured by the pure αP ansatz). The closed-form α upper bound Eq. (3) (correct algebra for single-feature L^4, but overshoots because it omits multi-feature interference). "Embedded $\equiv$ rotated plain" (matches on everything measured, but Procrustes-alignment of full Gram matrices was not done).

Flagged open: Whether α truly saturates at fixed F/n; whether the tight frame the network finds is unique across seeds; whether a refined three-parameter ansatz $R = \alpha P + \beta\, u\mathbf{1}^\top$ +skew would close the gap to trained R exactly.

7 Main gap worth closing

The single-feature L^4 ansatz (Eq. 3) systematically overpredicts α. Deriving a multi-feature correction, by integrating the ReLU-gated two-active-feature contribution, is tractable and would both sharpen the α prediction and likely explain the apparent saturation of α at fixed expansion ratio F/n for large F. I did not attempt this here.